

Eine Woche, ein Beispiel

7.7 special irreducible representations of simple Lie algebras

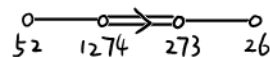
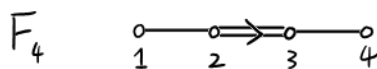
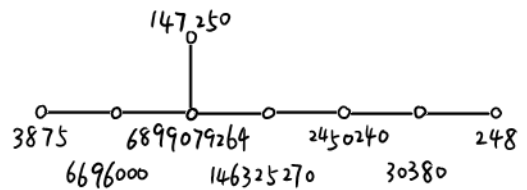
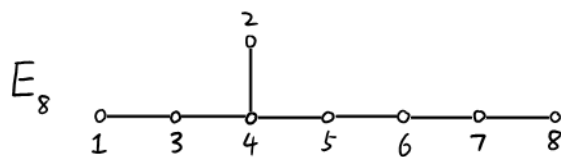
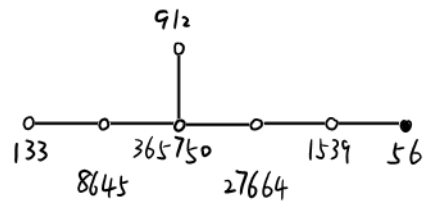
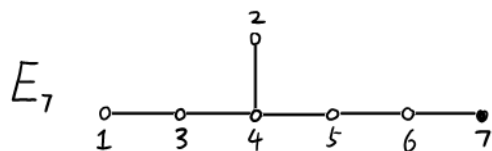
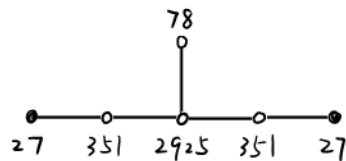
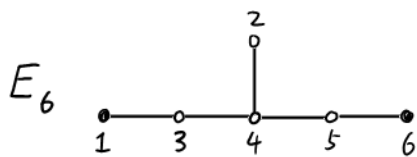
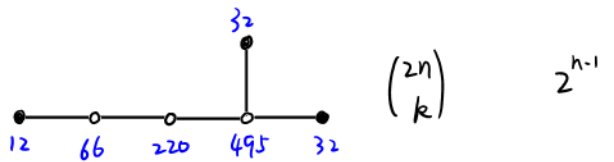
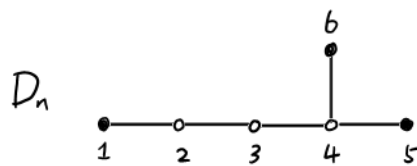
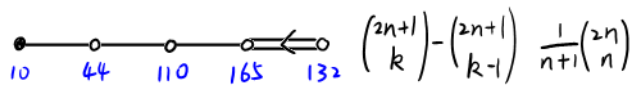
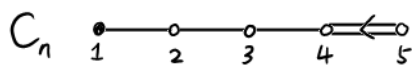
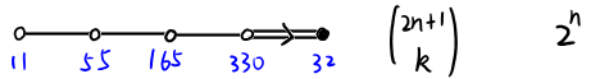
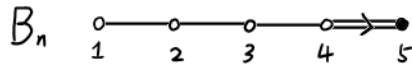
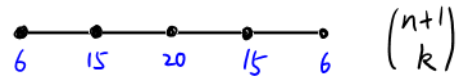
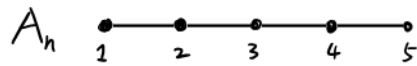
This document is a continuation for [2021.05.07_liegroup], [2021.07.18_irr_rep_of_semi_Lie_alg], [2021.07.25_irr_rep_of_SnAn], [2024.06.30_starting_functions].

The goal is to collect enough information on the representation sides, and then verify it on the perverse sheaf side.

Setting: We consider simple Lie algebras over $\mathbb{C}$.

1. labeling & basic rep dim
2. quasi-minuscule & adjoint reps

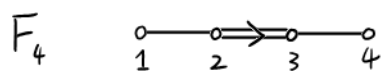
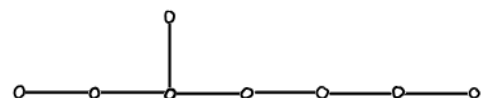
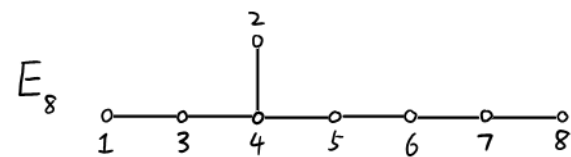
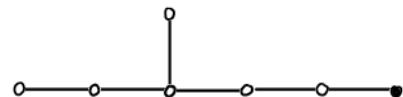
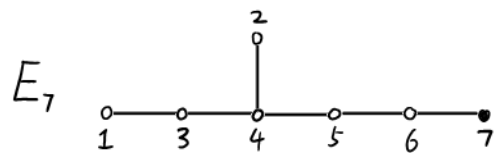
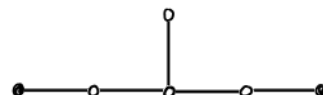
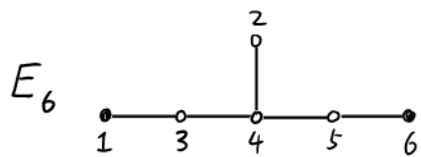
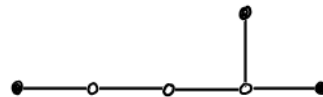
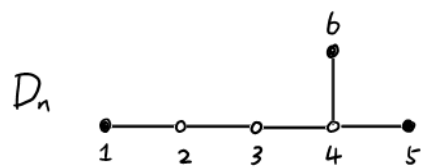
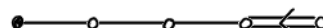
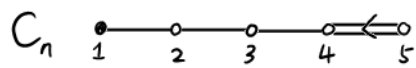
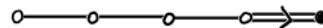
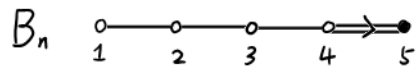
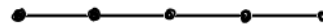
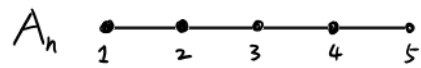
1. labeling & basic rep dim



long roots $\rightarrow$ short roots
 long weights $\rightarrow$ short weights
 short coroots $\rightarrow$ long coroots
 short coweights $\rightarrow$ long coweights

1. labeling & basic rep dim

This page is left for copy and paste.



2. quasi-minuscule & adjoint reps

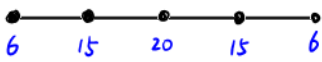
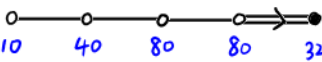
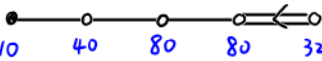
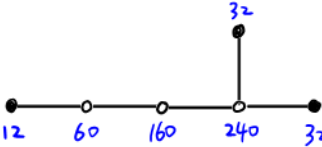
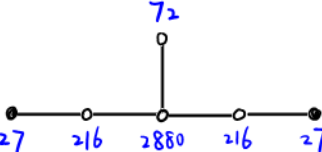
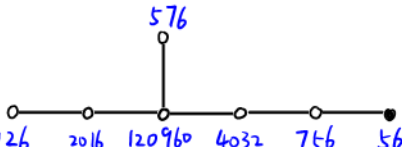
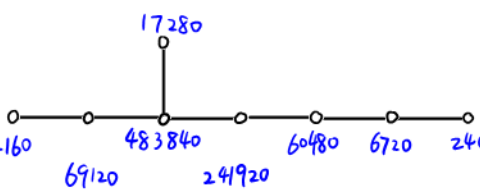
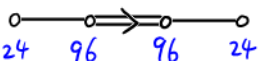
- minuscule rep
- x: quasi-minuscule rep
- x: adjoint rep

A_n		$\dim/\dim$ n^2+2n	rk/rk n
B_n		$2n^2+n/2n+1$	$n/1$
C_n		$2n^2+n/2n^2-n-1$	$n/n-1$
D_n		$2n^2-n$	n
E_6		78	6
E_7		133	7
E_8		248	8
F_4		$52/26$	$4/2$
G_2		$14/7$	$2/1$
A_1		3	1
B_2		$10/5$	$2/1$
D_3		15	3

quasi-minuscule and adjoint reps are exquisite, in the sense that their weights have only 2 or 3 orbits, and one of the orbit is the trivial weight. When the diagram is not simply-laced, the quasi-minuscule rep has orbits consisting of short roots and origin, while the adjoint rep has orbits consisting of long roots, short roots and origin.

3. orbit of weights

this can be easily computed from the order of Weyl groups. But I still want to see it very quickly, since they corresponds to the degree of some varieties.

A_n		$ W $ $(n+1)! = n! \times (n+1)$
B_n		$n! \times 2^n$
C_n		$n! \times 2^n$
D_n		$n! \times 2^{n-1}$
E_6		$51840 = 6! \times 72$
E_7		$2903040 = 7! \times 576$
E_8		$696729600 = 8! \times 17280$
F_4		$1152 = 4! \times 48$
G_2		$12 = 2! \times 6$